OVB, Interactions, and the Bridge to OLS

What omitting a variable does to the BLP

Gov 2001 · Scott Cunningham · Spring 2026

Frisch-Waugh-Lovell (FWL) Step 1: regress X_1 on X_2 and save the prediction error

Auxiliary regression:

$$X_1 = \pi_0 + \pi_1 X_2$$

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Prediction error — the part of X_1 that X_2 cannot explain:

$$\tilde{X}_1 \ = \ X_1 - \hat{X}_1$$

$\tilde{X}_1 \perp X_2$ by construction of the BLP

FWL Step 2: regress Y on $\tilde{X}_1$ — the slope equals β_1 from the long BLP

Bivariate BLP of Y on $\tilde{X}_1$:

$$Y = \gamma_0 + \gamma_1 \tilde{X}_1 \qquad \gamma_1 = \frac{\text{Cov}(Y, \tilde{X}_1)}{\text{Var}(\tilde{X}_1)}$$

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$$\boxed{\gamma_1 = \beta_1}$$

The Frisch-Waugh-Lovell theorem: the bivariate slope on $\tilde{X}_1$ equals the partial slope β_1 from the full multivariate BLP

Duncan (1961): does income or education drive occupational prestige?

45 US occupations, 1950 census

- Prestige: % rating occupation "good" or better
- Income: % earning > \$3,500/year
- Education: % with high school diploma
- Cor(Income, Education) = 0.73

Long BLP:

$$\widehat{\text{Prestige}} = -6.07 + \underbrace{0.599}_{\beta_{\text{inc}}}\ \text{Inc} + 0.546\,\text{Educ}$$

$$R^2 = 0.828$$

But the bivariate BLP says something different:

$$\widehat{\text{Prestige}} = 2.46 + \underbrace{1.080}_{\delta_{\text{inc}}}\ \text{Inc}$$

- Bivariate: $\delta_{\text{inc}} = 1.08$
- Long BLP: $\beta_{\text{inc}} = 0.60$
- Difference: income and education are correlated

Which slope is "right" for income?
FWL answers: $\beta_{\text{inc}} = 0.60$ —
the effect holding education fixed.

FWL in action: two regressions, one answer

Step 1: Regress Income on Education

$$\widehat{\text{Inc}} = 10.60 + 0.595\,\text{Educ}$$

Residual = income not explained by education:

$$\tilde{\text{Inc}} = \text{Inc} - \widehat{\text{Inc}}$$

By construction: $\text{Cor}(\tilde{\text{Inc}},\ \text{Educ}) = 0$

Step 2: Regress Prestige on $\tilde{\text{Inc}}$

$$\widehat{\text{Prestige}} = 47.69 + 0.599\,\widetilde{\text{Inc}}$$

The slope on $\tilde{\text{Inc}}$ is the part of income orthogonal to education.

Step 2 slope = 0.599 = β_{inc} from the long BLP (exact to 4 decimal places)

FWL is not an approximation. It is an algebraic identity.

Three regressions, two distinct slopes — FWL explains the difference

	Bivariate	**Long BLP**	**FWL**
	Prestige ~ Inc	Prestige ~ Inc + Educ	Prestige ~ $\widetilde{\text{Inc}}$
$\hat{\beta}_{\text{income}}$	**1.080**	**0.599**	**0.599**
$\hat{\beta}_{\text{education}}$	—	0.546	—
R^2	0.708	0.828	0.102

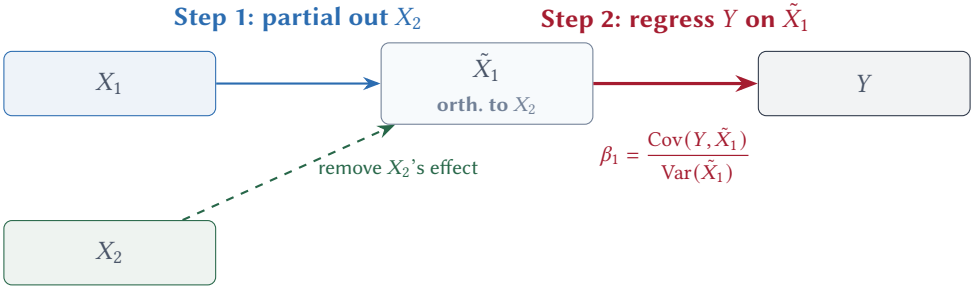
Why bivariate ≠ long:

- Income and education are correlated ($r = 0.73$)
- Bivariate income slope picks up education's effect too
- Long BLP separates them — income holding education fixed

Why FWL = long:

- $\widetilde{\text{Inc}}$ is orthogonal to education by construction
- Regressing on $\widetilde{\text{Inc}}$ is the same as controlling for education
- Coefficients are algebraically identical

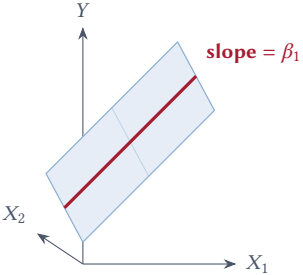
Frisch-Waugh-Lovell generalizes: every coefficient in the multivariate BLP is a partial-out



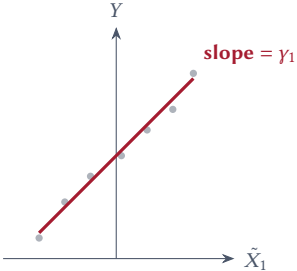
Every coefficient in a multivariate BLP is the result of this process for its own variable — simultaneously, not sequentially

FWL: the red line in the plane and the red line in the bivariate BLP are the same slope

Long BLP: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$



FWL bivariate BLP: $Y = \gamma_0 + \gamma_1 \tilde{X}_1$



Both red lines: same slope. Different regressions, same answer.

$\beta_1 = \gamma_1$: different regression, identical slope — the Frisch-Waugh-Lovell theorem

Long BLP

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

slope on $X_1 = \beta_1$

$=$

FWL bivariate BLP

$$Y = \gamma_0 + \gamma_1 \tilde{X}_1$$

slope on $\tilde{X}_1 = \gamma_1$

$\beta_1 = \gamma_1$ (e.g., vote share example: both = 1)

FWL wraps up multiple regression — now: what is U, and what happens when you omit a variable?

The BLP always comes with a prediction error:

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2}_{\text{BLP}(X_1, X_2)} + U$$

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U is the population prediction error

$U = Y - \text{BLP}(X_1, X_2)$

Always exists. Always defined.

FOC $\Rightarrow$ tautologies:

$\mathbb{E}[U] = 0$

$\text{Cov}(X_1, U) = 0$

$\text{Cov}(X_2, U) = 0$

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These are **not** assumptions — they hold by construction for every joint distribution of (Y, X_1, X_2)

U in BLP world vs. ε in structural world — same symbol, completely different meaning

BLP world: U

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + U$

U = population prediction error

$\text{Cov}(X, U) = 0$ is a **tautology**

Cannot fail — U is defined to be orthogonal to X

Structural world: ε

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

ε = unobserved causes of Y

$\text{Cov}(X, \varepsilon) = 0$ is an **assumption**

Can fail — this is what identification is about

OVB is **not** about $\text{Cov}(X, U) \neq 0$ — that can never happen in BLP world. OVB is about two different population objects answering different questions.

Frisch-Waugh-Lovell recovers β_1 from the long model — what if we omit Z entirely?

Frisch-Waugh-Lovell (FWL) (last lecture):

Two ways to get β_1 from the *long* BLP

Direct regression and partialing out give the same answer

New question:

What does the *short* BLP give?

$$m_S(X) = \delta_0 + \delta_1 X$$

Z omitted entirely — not partialed out, just gone

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The question: does omitting Z change the slope? By how much?

The OVB formula answers this exactly — pure algebra, no simulation needed

The short and long BLPs are two different population objects

Short BLP (omits Z):

$$m_S(X) = \delta_0 + \delta_1 X$$

Estimand: $\delta_1 = \dfrac{\text{Cov}(X, Y)}{\text{Var}(X)}$

What we get from the bivariate projection

Long BLP (includes Z):

$$m_L(X, Z) = \beta_0 + \beta_1 X + \gamma Z$$

Estimand: β_1 from the multivariate FOC

What we get when we hold Z fixed

The short and long BLPs are two different population objects

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Auxiliary projection:

$$Z = \pi_0 + \pi_1 X + v \qquad \Rightarrow \qquad \pi_1 = \frac{\text{Cov}(X, Z)}{\text{Var}(X)}$$

π_1: co-movement of Z and X in the population

Derive the OVB formula: substitute the auxiliary regression into the long BLP

Long BLP: $Y = \beta_0 + \beta_1 X + \gamma Z$ γ = coefficient on Z in the long model

Auxiliary: $Z = \pi_0 + \pi_1 X + v, \quad v \perp X$

Substitute Z into the long BLP:

$$
\begin{aligned}
Y &= \beta_0 + \beta_1 X + \gamma(\pi_0 + \pi_1 X + v) \\
&= \underbrace{(\beta_0 + \gamma\pi_0)}_{\text{new intercept}} + \underbrace{(\beta_1 + \gamma\pi_1)}_{\delta_1} X + \gamma v
\end{aligned}
$$

$\gamma v \perp X$ by construction: the short BLP slope picks up $\beta_1 + \gamma\pi_1$

$$\boxed{\delta_1 = \beta_1 + \pi_1 \cdot \gamma}$$

OVB formula: the short coefficient equals the long coefficient plus a term

$$\underbrace{\delta_1}_{\text{short BLP}} = \underbrace{\beta_1}_{\text{long BLP}} + \underbrace{\pi_1}_{\substack{\text{regression of} \\ Z \text{ on } X}} \cdot \underbrace{\gamma}_{\substack{\text{long coeff.} \\ \text{on } Z}}$$

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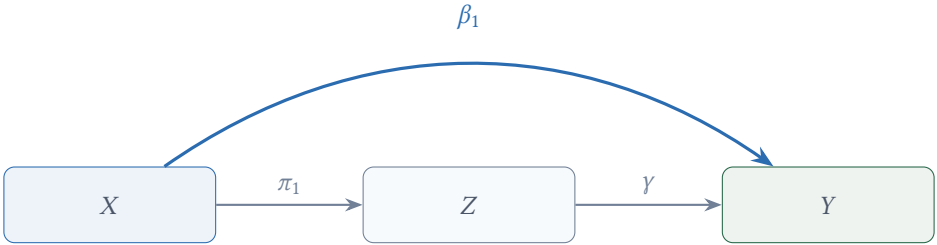
This is pure algebra

- Both δ_1 and β_1 are population BLP coefficients
- δ_1, β_1: population objects — no data needed
- No estimation, no sampling error

The term $\pi_1 \cdot \gamma$

- $\gamma \neq 0$: Z predicts Y conditional on X
- $\pi_1 \neq 0$: Z is correlated with X
- Both required: $\delta_1 = \beta_1$ if either is zero

OVB is zero under two conditions — each kills one channel



$\pi_1 \cdot \gamma = 0$ iff $\pi_1 = 0$ **or** $\gamma = 0$

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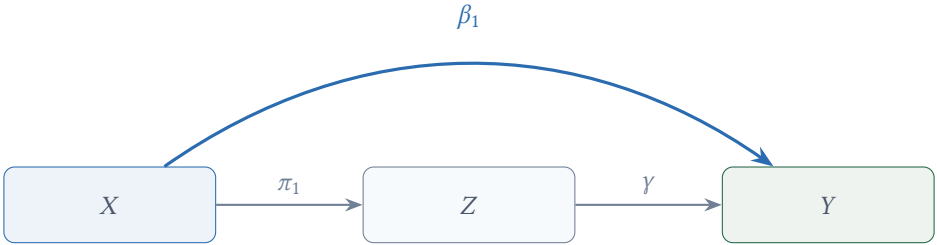

$\pi_1 \cdot \gamma = 0$ iff $\pi_1 = 0$ **or** $\gamma = 0$

$\gamma = 0$: Z does not predict Y once X is in the model

$\pi_1 = 0$: Z is orthogonal to X in the population
$\delta_1 = \beta_1$

Calling $\delta_1 - \beta_1$ "bias" is a causal claim — not a statistical one

What the OVB formula says:

$$\delta_1 - \beta_1 = \pi_1 \cdot \gamma$$

$\delta_1 \neq \beta_1$: two different population BLP coefficients answering different questions

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What "bias" requires:

- β_1 is the *true causal effect* of X
- δ_1 is a *contaminated* version
- Neither follows from algebra alone

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What makes β_1 causal:

$$(Y_{0i}, Y_{1i}) \perp X \mid Z \textbf{ (CIA)}$$

Requires a design argument — not more algebra

Today: $\delta_1 \neq \beta_1$ is algebra. Causality is a separate lecture.

Adding an interaction term changes the estimand — the slope on X_1 becomes a function of X_2

Additive BLP:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Slope on X_1: **constant**

$$\frac{\partial \text{BLP}}{\partial X_1} = \beta_1$$

Same effect of X_1 at every level of X_2

Interaction BLP:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

Slope on X_1: **a function of** X_2

$$\frac{\partial \text{BLP}}{\partial X_1} = \beta_1 + \beta_3 X_2$$

Effect of X_1 depends on level of X_2

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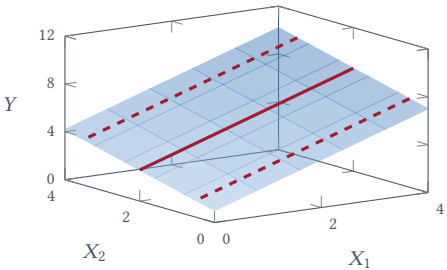
$$\frac{\partial \text{BLP}}{\partial X_1} = \beta_1 + \beta_3 X_2$$

Effect of X_1 depends on level of X_2

β_3: how much the effect of X_1 changes per unit increase in X_2

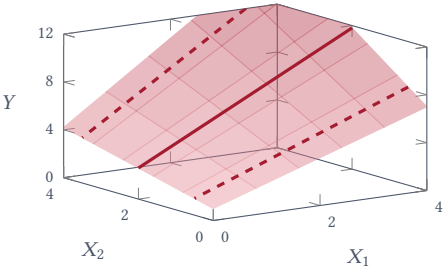
The additive BLP is a flat plane — the interaction BLP is a twisted surface

Additive: flat plane



All red lines: slope = $\beta_1 = 1.5$

Interaction: twisted surface



Red lines fan out: slope = $\beta_1 + \beta_3 X_2$

At $X_2 = 0$: the interaction slope equals β_1 — centering X_2 makes β_1 the average effect

Interaction model: $\dfrac{\partial \text{BLP}}{\partial X_1} = \beta_1 + \beta_3 X_2$

β_1 is the slope when $X_2 = 0$

If 0 is not in the support of X_2: β_1 has no direct interpretation

Center X_2: then β_1 = effect of X_1 at the mean of X_2

OVB from omitting $X_1 X_2$:

Using our earlier result: the bias on β_1 from omitting the interaction is $\beta_3 \cdot \pi_1$

$\pi_1 = 0$ when X_2 is mean-zero (centered)

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Center your covariates before interacting: β_1 in the additive model = β_1 in the interaction model

Saturated model: when $X_1, X_2 \in \{0, 1\}$, the interaction BLP *is* the CEF

Four cells, four parameters:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + U$$

	$X_2 = 0$	$X_2 = 1$
$X_1 = 0$	β_0	$\beta_0 + \beta_2$
$X_1 = 1$	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

One parameter per cell

BLP = CEF exactly

4 parameters, 4 cells
No approximation error

$\mathbb{E}[U \mid X_1, X_2] = 0$ everywhere

Saturated model: when $X_1, X_2 \in \{0, 1\}$, the interaction BLP *is* the CEF

Four cells, four parameters:

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$X_1 = 0$	β_0	$\beta_0 + \beta_2$
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One parameter per cell

BLP = CEF exactly

4 parameters, 4 cells
No approximation error
$\mathbb{E}[U \mid X_1, X_2] = 0$ everywhere

For binary covariates: every regression model with the full interaction is saturated — the CEF is linear, not approximately linear

Everything so far is a population object — OLS is the sample analog

Estimands covered in Lectures 10–11a:

- **Multivariate BLP:** $\beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$ — partial slopes, FWL
- **OVB formula:** $\delta_1 = \beta_1 + \pi_1 \cdot \gamma$ — pure algebra
- **Interaction BLP:** $\partial \text{BLP} / \partial X_1 = \beta_1 + \beta_3 X_2$ — effect heterogeneity
- **Saturated model:** BLP = CEF exactly for binary X

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Population
$\beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$
multivariate BLP

$\xrightarrow{\text{Lecture 11b}}$

sample analog:
$(X'X)^{-1}X'Y$

Sample
$\hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_k X_k$
OLS